1. Clinical Question & Core Hypothesis

Clinical Question

NMIBC prognosis reflects both histologic aggressiveness and the local anti-tumor immune response, but some morphologically high■risk or TIL■rich tumors still recur or progress — why does immune infiltration sometimes fail to prevent progression? 2) Tier 1 paths used: (a) tumor proliferative burden (MKI67) vs. immune effector capacity (CD8A) — the key mechanistic axis where high proliferation may outpace or blunt CD8+ T■cell control; (b) histomorphologic risk (Stage/Grade/CIS/budding) that signals invasion potential and microenvironment remodeling that can favor immune escape. I built on these because they directly address when/why immune presence (imaging TILs) does not translate to tumor control. 3) Tier 2 supporting evidence: cytotoxic/effector transcripts (IFNG, GZMB) and chemokines (CXCL9, CXCL10) indicate functional anti-tumor immunity, while checkpoints (PDCD1/CD274) and tumor architectural features (CIS, budding) can modulate effectiveness of CD8 responses; these support the core mechanistic interaction but are secondary modifiers. 4) Endpoint choice: PFI (progression-free interval) was selected because the clinical problem motivating NMIBC risk stratification is recurrence/progression to muscle■invasive disease rather than overall survival in early-stage NMIBC; PFI aligns with the decision points for surveillance and adjuvant therapy. 5) Deliberate exclusions: I excluded adding more than two anchor genes in the core hypothesis to keep the mechanistic claim sharp (other immune/cytotoxic/ checkpoint genes are relegated to secondary_modifiers). I also excluded alternative imaging features (TLS-like aggregates or TSR) to focus on the most directly relevant, widely assessable H&E metric — TIL density?

Working Hypothesis

Refined hypothesis: In NMIBC, the protective association between tumor-infiltrating lymphocytes (TIL density) and progression-free interval is context-dependent on tumor proliferative burden and CD8+ T-cell functional status. Specifically, high TIL density predicts longer PFI when MKI67 (proliferation) is low and CD8A (CD8+ cytotoxic cells) expression is high, reflecting effective cytotoxic control; conversely, when MKI67 is high and/or CD8A is low, high TIL density does not confer protection and may correlate with shorter PFI due to proliferation-driven immune evasion or T-cell dysfunction. This context dependence can be captured by an integrated NMIBC-IMES (Immune–Proliferation–Effector Score) that combines: TIL density (imaging feature), MKI67, CD8A, and functional/chemokine markers (IFNG, GZMB, CXCL9, CXCL10) with checkpoint indicators (PDCD1, CD274). Primary aim: test interactions between TIL density and (i) MKI67 and (ii) CD8A on PFI; secondary aims: assess whether functional markers and checkpoint expression modify these interactions. The study will classify tumors into risk strata with distinct PFI trajectories and explore potential for guiding intensified surveillance or adjuvant therapy in NMIBC.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MKI67, CD8A
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MKI67: MKI67) vs.

CD8A: CD8A) — the key mechanistic axis where high proliferation may outpace or blunt CD8+ T■cell control; (b) histomorphologic risk (Stage/Grade/CIS/budding) that signals invasion potential and microenvironment remodeling that can favor immune escape.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	IFNG	Supporting modifier from knowledge graph
Secondary	GZMB	Supporting modifier from knowledge graph
Secondary	CXCL9	Supporting modifier from knowledge graph
Secondary	CXCL10	Supporting modifier from knowledge graph
Secondary	PDCD1	Supporting modifier from knowledge graph
Secondary	CD274	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: reflect active anti-tumor responses, while PDCD1 and CD274 mark T-cell exhaustion and inhibitory signaling. An integrated score that merges these dimensions can stratify NMIBC risk beyond histology alone.

Prediction 2: , and Cox models with interaction terms to test the primary hypothesis. External validation in an independent NMIBC cohort would strengthen clinical translation.

Prediction 3: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- IFNG: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- GZMB: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.